

COURSE: RESEARCH METHOD-II

RESEARCH DESIGN

What is research design?

A research design is the plan or framework used to conduct a research study. It involves outlining the overall approach and methods that will be used to collect and analyze data in order to answer research questions or test hypotheses.

Purpose of research design?

- To provide a plan in answering research questioning.
- Each design has its own applicability depending on problem and objective of study.

Importance:

- **Clear direction:** the main role is providing direction to the research and guiding the choice of clear objectives, which help the researcher to focus on the specific research questions or hypotheses they want to investigate.
- **Control:** researchers can control variables, identify potential confounding factors, and use randomization to minimize bias and increase the reliability of their findings.
- **Replication:** Research designs provide the opportunity for replication. This helps to confirm the findings of a study and ensures that the results are not due to chance or other factors. Thus, a well-chosen research design also eliminates bias and errors.
- **Reliability:** Benefits of research design also include reducing inaccuracies and ensuring the reliability of the research (i.e., consistency of the research results over time, across different samples, and under different conditions).
- **Efficiency:** A strong research design helps increase the efficiency of the research process. Researchers can use a variety of designs to investigate their research questions, choose the most appropriate research design for their study, and use statistical analysis to make the most of their data. By effectively describing the data necessary for an adequate test of the hypotheses and explaining how such data will be obtained, research design saves a researcher's time.

Validity: A research design ensures the validity of the research, i.e., whether the results truly reflect the phenomenon being investigated.

Threats to validity:

- **History:** it refers to the events that may occur during the time frame of the study which are not actually a part of study. But they produce effects that influence the results of study either increasing or decreasing the expected results.

- **Selection:** it occurs when respondents of the study are chosen not only individual but as a group.
- **Testing:** it refers to the pre-test given that results in an improved performance in the post-test. To avoid this threat pre-test may not be needed to administration. However, if the pre-test is given, another measure is recommended to use as an alternate form of instruments.
- **Instrumentation:** it refers to unreliability in measuring instruments that may result to an invalid measurement of performance. The change in instruments used between the pretest and the post-test may result in an effect not caused by a treatment introduced.
- **Maturation:** this factor refers to the physiological and psychological changes that may happen to respondent of the study over a period of time. If the time frame of training program is quite long and rigid the participant may experience some psychological discomfort due to boredom, tiredness, hunger and like that.
- **Mortality:** it refers to the loss of participants during the post-test stage or even during the implementation of the study. When the same group of individuals is studied over a long period of time. By the time a follow-up study is conducted on the same group, some member may have dropped out or may refuse to cooperate future in study.

Different types of research design

Broadly, research design types can be divided into qualitative and quantitative research.

[Qualitative research](#) is subjective and exploratory. It determines relationships between collected data and observations. It is usually carried out through interviews with open-ended questions, observations that are described in words, etc.

[Quantitative research](#) is objective and employs statistical approaches. It establishes the cause-and-effect relationship among variables using different statistical and computational methods. This type of research is usually done using surveys and experiments.

Qualitative research vs. Quantitative research

Qualitative research	Quantitative research
Deals with subjective aspects, e.g., experiences, beliefs, perspectives, and concepts.	Measures different types of variables and describes frequencies, averages, correlations, etc.

Deals with non-numerical data, such as words, images, and observations.	Tests hypotheses about relationships between variables. Results are presented numerically and statistically.
In qualitative research design, data are collected via direct observations, interviews, focus groups, and naturally occurring data. Methods for conducting qualitative research are grounded theory, thematic analysis, and discourse analysis.	Quantitative research design is empirical. Data collection methods involved are experiments, surveys, and observations expressed in numbers. The research design categories under this are descriptive, experimental, correlational, diagnostic, and explanatory.
Data analysis involves interpretation and narrative analysis.	Data analysis involves statistical analysis and hypothesis testing.
The reasoning used to synthesize data is inductive.	The reasoning used to synthesize data is deductive.
Typically used in fields such as sociology, linguistics, and anthropology.	Typically used in fields such as economics, ecology, statistics, and medicine.
Example: Focus group discussions with women farmers about climate change perception.	Example: Testing the effectiveness of a new treatment for insomnia.

QUANTATIVE RESEARCH DESIGN

Survey designs: are procedures in quantitative research in which researcher administer a survey to a sample or to the entire population of people to describe the attitudes, opinion, behaviors, or characteristics of population. In this procedure, survey researchers collect quantitatively, numbers data using questionnaires (e.g. mailed questionnaires) or interviews and statistically analyze the data to describe trend about responses to questions and to test research questions or hypothesis.

When do we use survey research?

- to describe trends such as community interest in school bond issues or state or national trends about mandatory student policies?
- To determine individual opinions about policy, issues, such as whether students need a choice of school to attend
- Survey help identify important beliefs and attitudes of individuals, such as college students' beliefs about what constitutes abusive behaviors in dating relationships
- Survey as provide useful information to evaluate programs in schools such as the success of a robotics program in science education.

Nature, goals: The goal of a survey is to collect data from a representative sample of a population to draw conclusions about that larger population **basic step in survey research design:**

step1: decide if a survey is the best design to use; you need to decide whether survey research is the best design to use in the study. Surveys help to describe the trends in population or describe the relationship among variables or compare groups. Examples where surveys are most suitable are to access trends or characteristics of a population; learn about individuals attitudes, beliefs and practices; evaluate the success or effectiveness of a program; or identify the needs of community.

Step 2: identify the research questions or hypothesis: You can address both research questions and hypotheses in a survey design. Surveys lend themselves to hypothesis testing because you will be studying a sample to draw inferences to a population. Forms of research questions or hypotheses are those that:

Describe the characteristics or trends of a population of people, such as the frequency of tobacco use among male high school students

◆ Compare groups in terms of specific attributes, such as a comparison of teachers and administrators about attitudes toward "in-service" learning days

◆ Relate two or more variables, such as a survey of teachers to relate "burnout" to number of years of teaching.

Step 3. Identify the Population, the Sampling Frame, and the Sample: The process of survey research begins with identifying the population. This step requires defining the population, determining the number of people in it, and assessing whether you can obtain a list of names (i.e., the sampling frame) for the sample. Also, the population may need to be stratified before sampling, so select characteristics of the population (e.g., males and females) are represented in the sample.

Once you have identified the target population and compiled a list of its members, you can select the sample, preferably using random sampling procedures. You will need to identify an adequate sample size, using a sampling error formula.

Step 4. Determine the Survey Design and Data Collection Procedures: The researcher must also determine if the survey study will be cross-sectional or longitudinal.. The decision to use a

longitudinal or cross-sectional design relates to the nature of the problem studied, access to participants, and the time available to the researchers for data collection.

Step 5. Develop or Locate an Instrument: You need an instrument to collect or measure the variables in a study. It is easier to locate an instrument than to develop one. Standards of reliability and construct validity need to be applied to scores from existing instruments before you select them for use. If a study addresses only a few variables, researchers can design their own instruments. A check for the reliability and validity of scores from this instrument during data analysis is most important.

Step 6. Administer the Instrument: This step is perhaps the most time-consuming phase of survey research. It involves seeking and obtaining permission to conduct the survey and using procedures for data gathering, such as training interviewers or preparing questionnaires for mailing. It requires continually following up to obtain a high response rate, checking for response bias if questionnaires are used, and preparing the data for analysis by coding the information from the instruments into a computer file.

Step 7. Analyze the Data to Address the Research Questions or Hypotheses: The data analysis procedures will reflect the types of research questions or hypotheses the researcher plans to address in the study. Analysis consists of noting response rates, checking for response bias, conducting descriptive analysis of all items, and then answering descriptive questions. It might also involve testing hypotheses or research questions using inferential statistics.

Step 8. Write the Report: You should write the survey study using a standard quantitative structure that consists of an introduction, the review of the literature, the methods, the results, and the discussion. Specify in the "Methods" section of the study detailed information about the survey procedures. Include in the "Discussion" section comments about the generalizability of the results to the population.

Survey methods

Written survey: Written administration of questionnaires can take several forms: The questionnaires may be administered to a group, they may be dropped off at a particular location, or they may be mailed to the respondents. Group administration is familiar to you if you have completed course evaluations in college. Group administration is a very efficient use of time and money and can have a very high response rate if attendance by group members is high, as in a class. Drop-off administration is often done by an organization, such as a church, that has many members who attend a particular location over some period of time but may not all be present at one time. Copies of the questionnaire are dropped off in a particular location to be picked up at some later time. Mail administration is familiar to all.

Mail surveys: mail surveys are quick and convenient. they are used to distribute selfadministered questionnaires that respondents fill out on their own.

- **Advantages:** they usually can be completed relatively quickly. Because they are selfadministered. Mail surveys also avoid the problem due to interviewer bias. It is best method for dealing with highly personal or embarrassing topic, especially when anonymity of respondents preserved.

- **Disadvantages:** cost of copying and mailing. Respondents will not be able to ask questions. Researcher has little control over order in which respondent answer the questions. Order of questions may affect how respondents answer the questions. Mail survey is low response rate that can result in response rate bias. Response rate refers to percentage of people who complete the survey. For example, 30 out of 100 people complete the survey the response rate is 30%. A low response rate indicates that there could be a response rate bias that threatens the representativeness of a sample. For example, respondents with literacy problems low education level vision problem may not complete

the survey. Often people randomly selected for a sample are too busy or not interested enough in the study to return a completed questionnaire.

Personal interviews: face to face conversation is called personal interview. it is used to collect survey data in which respondents are usually contacted in their homes or in a shopping mall.

- **Advantages:** Personal interview allows greater flexibility in asking questions than does the mail survey. During interview the respondent can obtain clarification when question is unclear and trained interviewer can follow up incomplete or ambiguous answer to open ended questions. Interviewer control the order of questions and ensure that all respondents complete the questions in same order. Response rate of personal interviews has been higher than mail survey
- **Disadvantages:** personal interview is costly. The use of trained interviewers is expensive in terms of both money and cost. Another disadvantage is interviewer bias. The interviewer should be neutral medium through which questions and answers are transmitted. Interviewer bias occur when the interviewer records only selected portion of respondents answer or tries to adjust wording of a question to fit the respondent. To reduce interviewer bias research can employ highly motivated, well-paid interviewers who are trained to follow question wording exactly, to record responses accurately, and to use follow-up questions judiciously. Interviewers should be closely supervised by director of survey project.

Telephonic interviews: it is used for brief surveys.

- **Advantages:** It also provides better access to dangerous neighborhood locked buildings and respondents available only during evening hours. Interviews can be completed more quickly when contact are made by phone and interviewers can be better supervised when all interviews are conducted from one location.
- **Disadvantages:** selection bias exist when respondents are limited to those who have telephones and problem of interviewer bias remains. There is a limit to how long respondents are willing to stay on the phone and individuals may respond differently when talking to faceless voice than they would to a personal interviewer. Individuals from high socioeconomic group are more likely to have multiple phone numbers and hence might be overrepresented in a survey based on random-digit dialing. People may forget response option offered by interviewer.

Internet survey: computer-based survey. it is easy to administer participants complete a questionnaire online and click on submit button to have their responses recorded. There is multiple software that help to record and summarize the responses.

- **advantages:** This method is efficient and millions of participants who vary in age, ethnicity and nationality can be contacted through a few keystrokes on a computer. Online questionnaires are paperless thus saving natural resources and copying costs. Participants may respond when it is convenient and do so without leaving the comfort of their home, office room.



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Disadvantages: potential for sample biases at present, there is no way to generate a random sample of internet users. Selection bias is present because sample consist of individuals who have internet access. Higher income group have more internet access. Lack of control over research environment is also major disadvantage. Because researcher is not present there is no easy way to determine if respondents have a clear understanding of instructions.

Survey research design types

The three types of survey design are the cross-sectional design, the successive independent samples design, and the longitudinal design.

Cross-sectional design: In the cross-sectional design, one or more samples are drawn from the population(s) at one time. Cross-sectional designs allow researchers to describe the characteristics of a population or the differences between two or more populations, and correlational findings from cross-sectional designs allow researchers to make predictions. In a cross-sectional survey design, the researcher collects data at one point in time. For example, when middle school children complete a survey about teasing, they are recording data about their present views. For example, a cross-sectional design was used in a nationwide study of Internet use among 1,100 teens aged 12–17 This design has the advantage of measuring current attitudes or practices.

Advantages: It also provides information in a short amount of time, such as the time required for administering the survey and collecting the information. Cross-sectional designs are ideally suited for the descriptive and predictive goals of survey research.

Successive independent samples design: A successive-independent-samples design draws multiple random samples from a population at one or more times. different samples of respondents from the population complete the survey over a time period. The successive independent samples design allows researchers to study changes in a population over time. For successive independent samples designs to be effective,

- the samples must be drawn from the same population, and must be equally representative of it. If the samples are not comparable, the changes between samples may be due to demographic characteristics rather than time.
- In addition, the questions must be asked in the same way so that responses can be compared directly. this design measures the differences in individual participants' responses over time.

For example, public opinion researchers frequently ask independent samples of Americans the extent to which they approve of the U.S. president (referred to as the president's "approval ratings"). Changes in approval ratings over time are used to characterize Americans' opinions of the president's actions **Limitation:**

The purpose of the successive independent samples design is to describe changes over time in the distribution of population characteristics, not to describe changes in

- individual respondents. Accordingly, the successive independent samples design is not always helpful in ferreting out reasons for observed changes.
- A problem with the successive independent samples design occurs when the samples drawn from the population are not comparable—that is, not equally representative of the population. Although sophisticated statistical procedures exist to help unravel the problems associated with noncomparable successive samples, the best solution is to avoid the problem by carefully selecting successive samples that represent the same population

Longitudinal design: In the longitudinal design, the same respondents are surveyed over time in order to examine changes in individual respondents. A longitudinal survey design involves the survey procedure of collecting data about trends with the same population.

Example: Heatherton, Keel, and their colleagues have used the longitudinal design to investigate changes in attitudes and behaviors related to eating during the transitions from college to early adulthood and from early adulthood to middle years.

Human development from birth to old age.

Types:

- **Trend Studies:** In some surveys, researchers aim to study changes within some general population over a period of time. This form of longitudinal research is called a trend study. Trend studies are longitudinal survey designs that involve identifying a population and examining changes within that population over time. A popular example of this design is the Gallup Poll, which is used during elections to monitor trends in the population of voters from the primary to the final election.
- **Cohort Studies:** A cohort study is a longitudinal survey design in which a researcher identifies a subpopulation based on some specific characteristic and then studies that subpopulation over time. For example, a cohort group of 18-year-olds is studied in the year 2001. Five years later (in 2006), a group of 23-year-olds is studied. (They may or may not be the same individuals studied in 2001.) Five years after that (in 2011), a group of 28-year-olds is studied. While the individuals studied each time might be different, they must have been 18 years old in the year 2001 to qualify as representatives of the cohort group.
- **Panel Studies:** A panel study is a longitudinal survey design in which the researcher examines the same people over time. The high school seniors studied in 1998 will be the same people studied in 2000, 1 year after graduation, and again in 2002, 2 years after graduation. One disadvantage of a panel design is that individuals may be difficult to locate, especially 2 years after graduating from high school. The advantage to this type of study, however, is that the individuals studied will be the same each time, allowing the researcher to determine actual changes in specific individuals.

Advantages: The longitudinal design has two important advantages.

- First, the investigator can determine the direction and extent of change for individual respondents. Also, because changes in each individual's responses are assessed, it's easier to investigate reasons for attitude or behavior changes.
- Second, the longitudinal design is the best survey design when a researcher wishes to assess the effect of some naturally occurring event **Limitations:**
 - One possible problem is that respondents may strive courageously to be consistent across surveys. This can be particularly troublesome if the study is designed to assess changes in respondents' attitudes! Although their attitudes have actually changed, people may report their original attitudes in an effort to appear consistent.
 - Another potential problem is that the initial survey may sensitize respondents to the issue under investigation. For example, consider a longitudinal design used to assess students' concern about crime on campus. Once the study starts, participants may pay more attention to crime reports than they normally would. You might recognize this as an illustration of reactive measurement—people behaving differently because they know they are participating in a study.

Questionnaires and Interviews: A questionnaire is a form used in a survey design that participants in a study complete and return to the researcher. The participant chooses answers to questions and supplies basic personal or demographic information. An interview survey, however, is a form on which the researcher records answers supplied by the participant in the study. The researcher asks a question from an interview guide, listens for answers or observes behavior, and records responses on the survey.

Mailed Questionnaires: A mailed questionnaire is a form of data collection in survey research in which the investigator mails a questionnaire to members of the sample. A mailed questionnaire is economical because it involves only duplication and mailing expenses.

Web-Based Surveys or Questionnaires: A Web-based questionnaire is a survey instrument for collecting data that is available on the computer. Several software programs are available for designing, gathering, and analyzing survey data with sample questions and forms.

One-on-One Interviews: In one-on-one interviewing in survey research, investigators conduct an interview with an individual in the sample and record responses to closed-ended questions. In one-on-one interviewing in survey research, investigators conduct an interview with an individual in the sample and record responses to closed-ended questions.

Focus Group Interviews: In quantitative focus group interviews in survey research, the researcher locates or develops a survey instrument, convenes a small group of people (typically a group of 4 to 6) who can answer the questions, and records their comments on the instrument. For example, this group might consist of parents who evaluate a new math or science curriculum in a school.

In telephone interview surveys, the researcher records the participants' comments to questions on instruments over the telephone. The researcher develops or locates an instrument, obtains the

telephone numbers of participants in the sample, conducts the telephone calls, and asks the participants to answer questions on the instrument.

Instrument Design: Designing good survey instruments is a challenging and complex process. When survey researchers design an instrument for data collection, they typically perform the following steps:

1. They write different types of questions. These include personal, attitudinal, and behavioral questions; sensitive questions; and closed- and open-ended questions.
2. They use strategies for good question construction. This includes using clear language, making sure the answer options do not overlap, and posing questions that are applicable to all participants.
3. They perform a pilot test of the questions. This consists of administering the instrument to a small number of individuals and making changes based on their feedback.

Personal, Attitudinal, and Behavioral Questions

Background questions or demographic questions assess the personal characteristics of individuals in your sample. These questions can be easy (i.e., gender) or difficult to answer (i.e., level of income). Here are some examples of background questions:

What is your age?

How many years of teaching have you completed? (end of school year)

A second group of questions relates to obtaining individual attitudes or opinions from individuals in your sample. For example, you might ask:

How much do you agree or disagree with this statement?

Most days I am enthusiastic about being a student.

Strongly agree

Agree

Neither agree or disagree

Disagree

Strongly disagree

A third group of questions can solicit information about the actual behavior of individuals in the sample. For example:

Did you take a semester off during any of your 4 years of college?

Yes

No

Sensitive Questions; Some surveys contain sensitive questions that must be developed and used with care.

Sensitive questions might have to do with:

Drug and alcohol use (e.g., use of cocaine)

Mental health issues (e.g., paranoid behavior)

Open-and Closed-Ended Questions:

In closed-ended questions in surveys, the researcher poses a question and provides preset response options for the participant. A closed-ended question might be:

There are many reasons why adults wish to get more education. What is your most important reason for coming to adult basic education classes? (Check one.)

To be able to help my children with their schoolwork

To get a better job

To improve myself

To get a high school equivalency diploma

Open-ended questions in a survey are questions for which researchers do not provide the response options; the participants provide their own responses to questions. In an open-ended question, the participant supplies an answer.

For example:

Why are you attending adult education classes?

semi-closed-ended questions in a survey. This type of question has all the advantages of open-and closed-ended questions. The technique is to ask a closed-ended question and then ask for additional responses in an open-ended question. For example:

There are many reasons why adults wish to further their education. What is your most important reason for coming to adult basic education classes? (Check one.)

To be able to help my children with their schoolwork

To get a better job

To improve myself

To get a high school equivalency diploma

Other (please comment)

Question Construction: Using good questions helps participants feel that they understand the question and can provide meaningful answers. Good questions are clear and unambiguous, and they do not confuse the participants. They also show respect for the participant by being sensitive

to gender, class, and cultural needs of participants. When you construct questions for a survey questionnaire or interview, fit the questions to answers, include suitable response options, and do not overlap. A review of these potential question construction problems and some solutions will provide guidance for survey development.

The question is unclear. This usually occurs because words are vague or imprecise. Identify the unclear or vague words and replace them with words understood by participants in the study.

There are multiple questions. Here, the question actually contains two or more questions, called a double- or triple-barreled question. Reduce the multiple questions to a single question.

The question is wordy. When the question is too long, cut out unnecessary words to simplify and shorten the question. Look for excessive use of prepositions (e.g., more than three) or qualifying statements that lengthen the question.

◆ The question is negatively worded or wordy. If the question contains one or more negatives, such as "should not," the meaning becomes unclear. Also, reword the question if it leads the participants to one particular stance or another (e.g., using the word "pro-life"). Restate or reword the question to eliminate negative connotations or leading words.

The question includes jargon. Jargon may not be familiar to all participants in a study. Eliminate the jargon and use words familiar to all participants.

There are overlapping responses. This may lead to confusion when answering a question. Make sure that the response options do not overlap by creating distinct options.

There are unbalanced response options. In this case, the responses may be unbalanced in (e.g., "very important") and end with an "extent" word (e.g., "to a little extent"), rather than a matching adjective (e.g., "not important"). Decide on a single response option and use it consistently for all response categories for a question.

There is a mismatch between the question and the answers. The responses may not match the "action" word used in the question. Identify the verb or adjective in the question that will be the basis for the response options and create options using this word. (E.g., if the question says "to what extent," the answer will say "a great extent.")

The question includes overly technical language. When this occurs, the respondent may not have the level of understanding needed to respond to the question. Simplify the question so that all individuals will know the meaning of the words and can respond to the question

EXPERIMENTAL RESEARCH DESIGN

D.EF: Experimental design refers to how participants are allocated to different groups in an experiment. Experimental research is best suited for explanatory research, where the goal of the study is to examine cause-effect relationships. It also works well for research that involves a relatively limited and well-defined set of independent variables that can either be manipulated or controlled. Experimental research can be conducted in laboratory or field settings. Laboratory experiments, conducted in laboratory (artificial) settings, tend to be high in internal validity, but this comes at the cost of low external validity (generalizability), because the artificial

(laboratory) setting in which the study is conducted may not reflect the real world. Field experiments, conducted in field settings such as in a real organization

Purpose: the purpose of research design is to determine the explanations for the causes and effects of a precisely defined problem. An explanation of the link between variables is the main goal of this kind of research. The statistical method used to understand the link between the study's many characteristics is known as correlational research. Experimental research is the most common type of research design for people working in the sciences and a variety of other fields. Experimental design is an efficient method of optimizing the experimental conditions for SPE to maximize the amount of useful information obtained with the minimum number of experiments. It provides a more efficient and complete optimization compared with the 'vary one factor at a time approach with other factors assigned fixed values. Unlike 'vary one factor at a time approaches, suitable experimental designs are able to accommodate factor interaction

Experiment Terminology

Treatment and control groups. In experimental research, some subjects are administered one or more experimental stimulus called a treatment (the treatment group) while other subjects are not given such a stimulus (the control group). The treatment may be considered successful if subjects in the treatment group rate more favorably on outcome variables than control group subjects. Multiple levels of experimental stimulus may be administered, in which case, there may be more than one treatment group. For example, in order to test the effects of a new drug intended to treat a certain medical condition like dementia, if a sample of dementia patients is randomly divided into three groups, with the first group receiving a high dosage of the drug, the second group receiving a low dosage, and the third group receives a placebo such as a sugar pill (control group), then the first two groups are experimental groups and the third group is a control group. After administering the drug for a period of time, if the condition of the experimental group subjects improved significantly more than the control group subjects, we can say that the drug is effective. We can also compare the conditions of the high and low dosage experimental groups to determine if the high dose is more effective than the low dose.

Treatment manipulation. Treatments are the unique feature of experimental research that sets this design apart from all other research methods. Treatment manipulation helps control for the "cause" in cause-effect relationships. Naturally, the validity of experimental research depends on how well the treatment was manipulated. Treatment manipulation must be checked using pretests and pilot tests prior to the experimental study. Any measurements conducted before the treatment is administered are called pretest measures, while those conducted after the treatment are posttest measures.

Random selection and assignment. Random selection is the process of randomly drawing a sample from a population or a sampling frame. Random assignment is however a process of randomly assigning subjects to experimental or control groups. This is a standard practice in true experimental research to ensure that treatment groups are similar (equivalent) to each other and to the control group, prior to treatment administration. Random selection is related to sampling, and is therefore, more closely related to the external validity (generalizability) of findings.

However, random assignment is related to design, and is therefore most related to internal validity. It is possible to have both random selection and random assignment in well-designed experimental research, but quasi-experimental research involves neither random selection nor random assignment.

- **Ecological validity:** The degree to which an investigation represents real-life experiences.
- **Experimenter effects:** These are the ways that the experimenter can accidentally influence the participant through their appearance or behavior.
- **Demand characteristics:** The clues in an experiment lead the participants to think they know what the researcher is looking for (e.g., the experimenter's body language).
- **Independent variable (IV):** The variable the experimenter manipulates (i.e., changes) is assumed to have a direct effect on the dependent variable.
- **Dependent variable (DV):** Variable the experimenter measures. This is the outcome (i.e., the result) of a study.
- **Extraneous variables (EV):** All variables which are not independent variables but could affect the results (DV) of the experiment. Extraneous variables should be controlled where possible.
- **Confounding variables:** Variable(s) that have affected the results (DV), apart from the IV. A confounding variable could be an extraneous variable that has not been controlled.
- **Random Allocation:** Randomly allocating participants to independent variable conditions means that all participants should have an equal chance of taking part in each condition.
- The principle of random allocation is to avoid bias in how the experiment is carried out and limit the effects of participant variables.
- **Order effects:** Changes in participants' performance due to their repeating the same or similar test more than once. Examples of order effects include:
 - (i) practice effect: an improvement in performance on a task due to repetition, for example, because of familiarity with the task;
 - (ii) fatigue effect: a decrease in performance of a task due to repetition, for example, because of boredom or tiredness.

Experimental method: control and variability,

Some of the characteristics are:

□ **Variables:** Dependent, independent, and extraneous variables are all present in experimental research. The dependent variables are the variables that are being treated or altered and are frequently referred to as the study subject. The experimental treatment being applied to the dependent variables is represented by the independent variables. Extraneous variables, on the other hand, are outside influences on the experiment that may potentially contribute to the change. Experimental research may include multiple independent variables, e.g. time, skills, test scores, etc.

□ **Setting:** The setting is the location of the experiment. Many experiments are conducted in the laboratory, where extraneous factors may be controlled and therefore eliminated. Other tests are conducted in a less controlled environment. The type of environment utilized in research is determined by the nature of the experiment.

□ **Control:** Except for the independent variable, the researcher must control all relevant variables. Control is extremely crucial in experimental research. Without control, it is impossible to deduce the effects of an independent variable. To comprehend the idea of control in experimentation, one must first comprehend the two fundamental laws that underlie experimental research.

□ **Manipulation:** A predetermined set of various situations is placed on the individuals chosen for the experiment during the manipulation procedure. The collection of circumstances is known as the independent variable, the experimental variable, or the treatment variable.

□ **Observation:** The experimenter is liable for observing changes in a dependent variable as a result of manipulating an independent variable. Process of experimental design Creating a set of methods to test a hypothesis systematically is referred to as experimental design. The design of an experiment involves multiple phases.

The steps of experimental research design are:

□ **Defining Research Problem:** Identifying the problem is the first step in any type of research. Experimental research is basically conducted to solve practical problems rather than theoretical problems or gaps.

□ **Determining variables:** A research question must be transformed into an experimental hypothesis by defining the key variables and predicting their relationships.

□ **Formulating Hypothesis:** Conducting the research requires creating a precise, testable hypothesis that corresponds to the research topic.

□ **Designing a controlled experiment:** Creating an experimental plan that addresses all the constituent elements, conditions, and connections between the outputs.

Designing a controlled experiment requires:

1. manipulating the independent variable(s) systematically and precisely
2. Measuring the dependent variable(s) with precision
3. Controlling any factors that can cause confusion

□ **Designing experimental treatments to manipulate the independent variable:** Manipulation of the independent variable can have an impact on the experiment's external validity or the extent to which the results can be extended and applied to the broader world. The first step is, to consider how widely the independent variable should be varied. The second step is, to decide how finely to alter the independent variable. This decision is sometimes made by the experimental system, but it is often required to make one, and this will affect how much the researcher may deduce from the data. If the subjects are exposed to more than one treatment then

the findings could only be generalized to individuals exposed to the same treatments in the same order of presentation.

□ **Measuring dependent variables** Finally, determining the collection method of data on the dependent variable outcomes. The aim should be to achieve accurate and valid measures that reduce research bias or inaccuracy. Some factors can be best measured using scientific tools. Others may need to be operationalized prior to being measured.

Types of experimental design:

Types of design include repeated measures, independent groups, and matched pairs designs. The researcher must decide how he/she will allocate their sample to the different experimental groups. For example, if there are 10 participants, will all 10 participants participate in both groups (e.g., repeated measures), or will the participants be split in half and take part in only one group each.

Independent measure designs/Between group design: Independent measures design, also known as between-groups, is an experimental design where different participants are used in each condition of the independent variable. This means that each condition of the experiment includes a different group of participants. This should be done by random allocation, ensuring that each participant has an equal chance of being assigned to one group. The most effective independent groups design is one that uses random assignment of subjects to conditions in order to form comparable groups prior to implementing the independent variable. When random assignment to conditions is used, the independent groups design is called a random groups design. The logic of the design is straightforward. The groups are formed so as to be similar on all important characteristics at the start of the experiment. Next, in the experiment itself, the groups are treated the same except for the level of the independent variable. Thus, any difference between the groups on the dependent variable must be caused by the independent variable.

Independent measures involve using two separate groups of participants, one in each condition.

For example: In the experiment small groups of young girls (5½–6½ years old) were read a story about “Mira” as she went shopping for clothes and prepared to go to a birthday party. As they heard the story, the girls looked at picture books with six scenes related to the story. In one condition of the experiment, the picture books had images of Barbie in the scenes of the story (e.g., shopping for a party outfit, getting ready for the party). In a second condition the picture books had similar scenes but the figure pictured was the “Emme” doll. The Emme fashion doll is an attractive doll with more realistic body proportions, representing a U.S. dress size 16 (see Figure 6.2). Finally, in the third condition of the experiment the picture books did not depict Barbie or Emme (or any body) but, instead, showed neutral images related to the story (e.g., windows of clothes shops, colorful balloons). These three versions of the picture books (Barbie, Emme, neutral) represent three levels of the independent variable that was manipulated in the experiment. Because different groups of girls participated in each level of the independent variable, the experiment is described as an independent groups design. Dittmar and her colleagues tested the hypothesis that only thin body ideals, as represented by Barbie, would cause body dissatisfaction. At the end of the story, the young girls turned in their picture books and completed a questionnaire designed for their age level.

- manipulated an independent variable that varied the images girls viewed,
- eliminated other plausible explanations through holding relevant conditions constant, and
- balanced individual differences among the groups through random assignment to conditions.

Independent Variable (IV) = Sleep		
	1. 2 hours per night	2. 10 hours per night
	Group A (10 students)	Group B (10 different students)
DV	Reaction Time	Reaction Time

Pro: Avoids order effects (such as practice or fatigue) as people participate in one condition only. If a person is involved in several conditions, they may become bored, tired, and fed up by the time they come to the second condition or become wise to the requirements of the experiment!

Con: More people are needed than with the repeated measures design (i.e., more timeconsuming).

Differences between participants in the groups may affect results, for example, variations in age, gender, or social background. These differences are known as participant variables (i.e., a type of extraneous variable).

Control: After the participants have been recruited, they should be randomly assigned to their groups. This should ensure the groups are similar, on average (reducing participant variables).

Block Randomization • Block randomization balances subject characteristics and potential confounding that occur during the time in which the experiment is conducted, and it creates groups of equal size. A common procedure for carrying out random assignment is block randomization. First, let us describe exactly how block randomization is carried out, and then we will look at what it accomplishes. Suppose we have an experiment with five conditions (labeled, for convenience, as A, B, C, D, and E). One “block” is made up of a random order of all five conditions:

One block of conditions → Random order of conditions

A B C D E

C A E B D

Repeated measure design/Within group design

Repeated Measures design is an experimental design where the same participants participate in each independent variable condition. This means that each experiment condition includes the same group of participants. Repeated Measures design is also known as within-groups or [withinsubjects design](#).

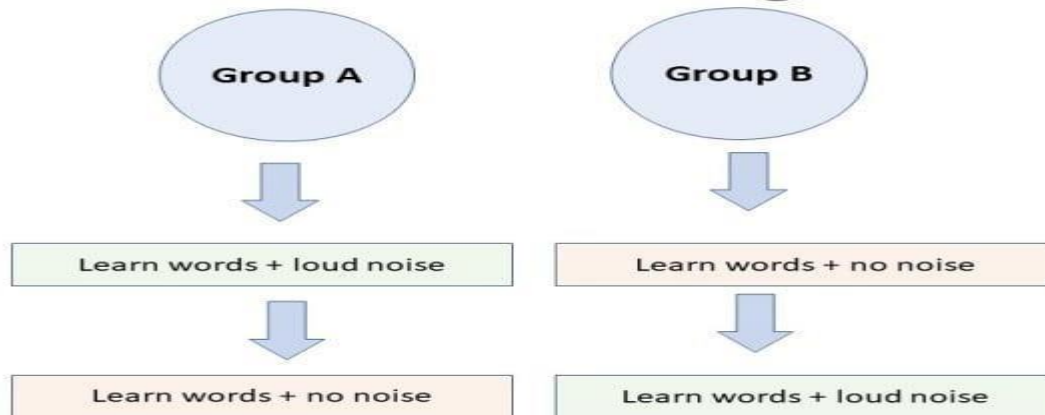
- **Pro:** As the same participants are used in each condition, participant variables (i.e., individual differences) are reduced. Fewer people are needed as they participate in all conditions (i.e., saves time).
- **Con:** There may be order effects. Order effects refer to the order of the conditions affecting the participants' behavior. Performance in the second condition may be better because the participants know what to do (i.e., practice effect). Or their performance might be worse in the second condition because they are tired (i.e., fatigue effect). This limitation can be controlled using counterbalancing.
- **Control:** To combat order effects, the researcher counter-balances the order of the conditions for the participants. Alternating the order in which participants perform in different conditions of an experiment.
- **Counterbalancing**

Suppose we used a repeated measures design in which all of the participants first learned words in “loud noise” and then learned them in “no noise.” We expect the participants to learn better in “no noise” because of order effects, such as practice. However, a researcher can control for order effects using counterbalancing.

The sample would be split into two groups: experimental (A) and control (B). For example, group 1 does ‘A’ then ‘B,’ and group 2 does ‘B’ then ‘A.’ This is to eliminate order effects.

Although order effects occur for each participant, they balance each other out in the results because they occur equally in both groups.

Counter Balancing



3. Matched Pairs Design

A matched pairs design is an experimental design where pairs of participants are matched in terms of key variables, such as age or socioeconomic status. One member of each pair is then placed into the experimental group and the other member into the control group. One member of each matched pair must be randomly assigned to the experimental group and the other to the control group. A matched groups design may be used to create comparable groups when there are too few subjects available for random assignment to work effectively. • Matching subjects on the dependent variable task is the best approach for creating matched groups, but performance on any matching task must correlate with the dependent variable task. • After subjects are matched on the matching task, they should then be randomly assigned to the conditions of the independent variable. The matched groups design is useful only when a good matching task is available.

Example: For example, if the dependent variable in the experiment is blood pressure, participants should be matched on blood pressure prior to the start of the experiment. The matching is accomplished by measuring the blood pressure of all participants and then forming pairs or triples or quadruples of participants (depending on the number of conditions in the experiment) who have identical or very similar blood pressures. Thus, at the start of the experiment, participants in the different groups will have, on average, equivalent blood pressure. Researchers can then reasonably attribute any group differences in blood pressure at the end of the study to the treatment (p

Independent Variable (IV) = Sleep		
	1. 2 hours per night	2. 10 hours per night
	Group A (10 students)	Group B (10 students, matched for age, gender, and normal sleeping length)
DV	Reaction Time	Reaction Time

resuming other potential variables have been held constant or balanced). **Pro**

- Reduces participant variables because the researcher has tried to pair up the participants so that each condition has people with similar abilities and characteristics.
- It avoids order effects, so counterbalancing is not necessary.

Con:

- Very time-consuming trying to find closely matched pairs.
- If one participant drops out, you lose 2 PPs' data.
- Impossible to match people exactly unless they are identical twins!

Control: Members of each pair should be randomly assigned to conditions. However, this does not solve all these problems.

matching is not sufficient to form comparable groups in an experiment. For example, consider a matched group design to compare two different methods of caring for premature infants so as to increase their body weight. Six pairs of premature infants could be matched on their initial body weight. There remain, however, potentially relevant characteristics of the participants beyond those measured by the matching task. For example, the two groups of premature infants may not be comparable in their general health or in their degree of parental attachment. It is important, therefore, to use random assignment in the matched groups design to try to balance other potential factors beyond the matching task. Specifically, after matching the infants on body weight, individuals in each pair would be randomly assigned to one of the two treatment groups. In conclusion, the matched groups design is a better alternative than the random groups design when a good matching task is available and when only a small number of subjects is available for an experiment that requires separate groups for each condition

Methodological issues in experimental research design:

Individual differences

Assessing conditions

QUASI- EXPERIMENTAL DESIGN

There are several common differences between true and quasi-experimental designs.

	True experimental design	Quasi-experimental design
Assignment to treatment	The researcher randomly assigns subjects to control and treatment groups.	Some other, non-random method is used to assign subjects to groups.
Control over treatment	The researcher usually designs the treatment .	The researcher often does not have control over the treatment , but instead studies preexisting groups that received different treatments after the fact.
Use of control groups	Requires the use of control and treatment groups .	Control groups are not required (although they are commonly used).

Quasi-experimental design: is a research method that seeks to evaluate the causal relationships between variables, but without the full control over the independent variable(s) that is available in a true experimental design. In a quasi-experimental design, the researcher uses an existing group of participants that is not randomly assigned to the experimental and control groups. Instead, the groups are selected based on pre-existing characteristics or conditions, such as age, gender, or the presence of a certain medical condition.

Example: Here are some examples of real-time quasi-experimental designs:

- **Evaluating the impact of a new teaching method:** In this study, a group of students are taught using a new teaching method, while another group is taught using the traditional method. The test scores of both groups are compared before and after the intervention to determine whether the new teaching method had a significant impact on student performance.
- **Assessing the effectiveness of a public health campaign:** In this study, a public health campaign is launched to promote healthy eating habits among a targeted population. The behavior of the population is compared before and after the campaign to determine whether the intervention had a significant impact on the target behavior.
- **Examining the impact of a new medication:** In this study, a group of patients is given a new medication, while another group is given a placebo. The outcomes of both groups are compared to determine whether the new medication had a significant impact on the targeted health condition.
- **Evaluating the effectiveness of a job training program:** In this study, a group of unemployed individuals is enrolled in a job training program, while another group is not enrolled in any program. The employment rates of both groups are compared before and after the intervention to determine whether the training program had a significant impact on the employment rates of the participants.

When to use Quasi-Experimental Design

Here are some situations where quasi-experimental designs may be appropriate:

- When the research question involves investigating the effectiveness of an intervention, policy, or program
- When the sample size is small
- When the research question involves investigating a naturally occurring event
- When the research question involves investigating a long-term intervention
- When the research question involves investigating the impact of a variable that cannot be manipulated
- Sometimes it would be unethical to provide or withhold a treatment on a random basis, so a true experiment is not feasible. In this case, a quasi-experiment can allow you to study the same causal relationship without the ethical issues. The Oregon Health Study is a good example. It would be unethical to randomly provide some people with health insurance but purposely prevent others from receiving it solely for the purposes of research.

Steps in Quasi-Experimental Design

Here are the general steps involved in conducting a quasi-experimental design:

- **Identify the research question:** Determine the research question and the variables that will be investigated.
- **Choose the design:** Choose the appropriate quasi-experimental design to address the research question. Examples include the pretest-posttest design, non-equivalent control group design, regression discontinuity design, and interrupted time series design.
- **Select the participants:** Select the participants who will be included in the study. Participants should be selected based on specific criteria relevant to the research question.
- **Measure the variables:** Measure the variables that are relevant to the research question.

This may involve using surveys, questionnaires, tests, or other measures.

- **Implement the intervention or treatment:** Implement the intervention or treatment to the participants in the intervention group. This may involve training, education, counseling, or other interventions.
- **Collect data:** Collect data on the dependent variable(s) before and after the intervention. Data collection may also include collecting data on other variables that may impact the dependent variable(s).
- **Analyze the data:** Analyze the data collected to determine whether the intervention had a significant impact on the dependent variable(s).
- **Draw conclusions:** Draw conclusions about the relationship between the independent and dependent variables. If the results suggest a causal relationship, then appropriate recommendations may be made based on the findings.

Applications of Quasi-Experimental Design

Here are some applications of quasi-experimental design:

- **Educational research:** Quasi-experimental designs are used to evaluate the effectiveness of educational interventions, such as new teaching methods, technology-based learning, or educational policies.
- **Health research:** Quasi-experimental designs are used to evaluate the effectiveness of health interventions, such as new medications, public health campaigns, or health policies.
- **Social science research:** Quasi-experimental designs are used to investigate the impact of social interventions, such as job training programs, welfare policies, or criminal justice programs.
- **Business research:** Quasi-experimental designs are used to evaluate the impact of business interventions, such as marketing campaigns, new products, or pricing strategies.
- **Environmental research:** Quasi-experimental designs are used to evaluate the impact of environmental interventions, such as conservation programs, pollution control policies, or renewable energy initiatives.

Types of quasi-experimental designs

Many types of quasi-experimental designs exist. Here we explain three of the most common types: nonequivalent groups design, regression discontinuity, and natural experiments.

Nonequivalent groups design

- In nonequivalent group design, the researcher chooses existing groups that appear similar, but where only one of the groups experiences the treatment.
- In a true experiment with random assignment, the control and treatment groups are considered equivalent in every way other than the treatment. But in a quasi-experiment where the groups are not random, they may differ in other ways—they are *nonequivalent groups*.
- When using this kind of design, researchers try to account for any confounding variables by controlling for them in their analysis or by choosing groups that are as similar as possible.
- This is the most common type of quasi-experimental design.

Example: Nonequivalent groups design You hypothesize that a new after-school program will lead to higher grades. You choose two similar groups of children who attend different schools, one of which implements the new program while the other does not.

By comparing the children who attend the program with those who do not, you can find out whether it has an impact on grades.

Regression discontinuity

- Many potential treatments that researchers wish to study are designed around an essentially arbitrary cutoff, where those above the threshold receive the treatment and those below it does not.
- Near this threshold, the differences between the two groups are often so minimal as to be nearly nonexistent. Therefore, researchers can use individuals just below the threshold as a control group and those just above as a treatment group.
- Example: Regression discontinuity Some high schools in the United States are set aside for high-achieving students, who must exceed a certain score on a test to be allowed to attend. Those who pass this test most likely differ systematically from those who do not. •
However, since the exact cutoff score is arbitrary, the students near the threshold—those who just barely pass the exam and those who fail by a very small margin—tend to be very similar, with the small differences in their scores mostly due to random chance. You can therefore conclude that any outcome differences must come from the school they attended.
- To test the impact of attending a selective school, you can study the long-term outcomes of these two groups of students (those who barely passed and those who barely failed).

Natural experiments

- In both laboratory and field experiments, researchers normally control which group the subjects are assigned to. In a natural experiment, an external event or situation (“nature”) results in the random or random-like assignment of subjects to the treatment group.
- Even though some use random assignments, natural experiments are not considered to be true experiments because they are observational in nature.
- Although the researchers have no control over the independent variable, they can exploit this event after the fact to study the effect of the treatment.
- Example: Natural experiment The Oregon Health Study is one of the most famous natural experiments. In 2008, the state of Oregon decided to expand enrollment in Medicaid, America’s low-income public health insurance program, to more low-income adults. •
However, as they could not afford to cover everyone who they deemed eligible for the program, they instead allocated spots in the program based on a random lottery.
- Researchers were able to study the impact of the program by using the enrolled individuals as a randomly assigned treatment group, and the others who were eligible but did not succeed in the lottery as a control group.

Pretest-posttest design

This design involves measuring the outcome variable before and after the treatment is applied. Participants are not randomly assigned to groups, and there is no control group. The focus is on comparing the change in the outcome variable within the same group over time.

Time series design

In this design, multiple measurements of the outcome variable are taken at regular intervals before, during, and after the treatment. The goal is to observe any patterns or trends in the data that can be attributed to the treatment.

Retrospective/Ex post Facto design: **Ex post facto design**, also known as "after-the-fact" research, is defined as a research method that looks into how an independent variable (groups with certain qualities that already exist prior to a study) affects a [dependent variable](#). This entails particular characteristics or traits of a participant that cannot be manipulated. Ex post facto design is considered a [quasi-experimental](#) type of study, which means that participants are not randomly assigned, but rather grouped together based upon specific characteristics or traits they share.

Ex post facto research focuses on how actions that have already occurred can predict certain causes. Therefore, a researcher cannot manipulate or modify actions or behaviors that have already occurred, or specific traits and characteristics a participant embodies. This type of research study tests hypotheses to locate a cause-and-effect relationship between the independent and dependent variables.

An example of ex post facto research is if a researcher wants to determine a cause-and-effect relationship between height and IQ scores. A researcher cannot physically alter someone's height, therefore specific groups will be put together based on how tall they are. One group may be put together if they are taller than 5'5, while another group may be put together if they are shorter than 5'5.

Prospective quasi-experimental design: refers to the likely prevalence of phenomenon, situation problem attitude that has work in present or outcome in the future. Experiments are usually classified as prospective studies as the researcher must wait for an intervention to register its effect on the study population.

Example, to establish the effects of counselling service on the extent of marital problems.

Event-Specific Alignment designs: Event-specific alignment in research refers to the deliberate structuring of an event—such as a conference, workshop, or seminar—to support and enhance specific research goals, objectives, and outcomes.

Here are key considerations for designing event-specific alignment in research:

Define Research Goals and Objectives: Clearly articulate the specific research goals, objectives, and hypotheses that the event aims to address or contribute to. This forms the foundation for aligning all aspects of the event.

Select Relevant Themes and Topics: Choose themes, topics, and sessions that are directly aligned with the research focus. Ensure that each session or activity contributes to advancing understanding or knowledge in the research area.

Curate Content and Speakers: Select speakers, presenters, and panelists whose expertise and contributions align closely with the research objectives. Their insights should complement and support the research themes being explored.

Design Interactive Sessions: Structure sessions and activities in ways that encourage interaction, discussion, and collaboration among participants. This fosters a productive environment for exchanging ideas and insights relevant to the research.

Integrate Data Collection Opportunities: Incorporate opportunities for data collection, such as surveys, interviews, or observations, within the event framework. Ensure that these methods align with the research methodology and ethics requirements.

Facilitate Networking and Collaboration: Create networking opportunities that facilitate connections among researchers, practitioners, and stakeholders interested in the research area. These connections can lead to collaborations and further research opportunities.

Align Logistics and Operations: Ensure logistical arrangements—including venue selection, scheduling, catering, and technical support—support the smooth operation of the event and contribute to a conducive research environment.

Promote Research Findings and Discussions: Provide platforms for presenting research findings, whether through presentations, posters, or discussions. Encourage participants to critically engage with and provide feedback on the research.

Evaluate Impact and Outcomes: Plan for evaluating the impact and outcomes of the event in relation to the research objectives. Collect feedback from participants, assess knowledge dissemination, and measure any changes in understanding or practices.

Communicate Results and Follow-Up: Communicate the outcomes and findings of the event to participants, stakeholders, and the broader research community. Consider follow-up activities or publications to further disseminate research contributions.

By meticulously aligning an event with research goals and objectives, researchers can maximize the event's effectiveness as a platform for knowledge exchange, collaboration, and advancing scientific inquiry in their field of study.

Twin studies and Adoption studies

Adoption studies: are research studies that compare an adopted person's behavior to both their biological and adoptive parents' behavior. An adopted person has many of the same **genetics**, or inherited traits as their biological parents, which may result in some of their behaviors. Although adoptive parents may not share any of the same genes with their children,

they can still influence their behavior. Just growing up under that adoptive parent's care can lead an adopted child to pick up similar behaviors. When a person spends a lot of time around a certain group or individual, they tend to pick up some of their behaviors. By comparing these two sets of parents, researchers can better compare the impact of genetics versus environmental factors on behaviors.

These studies are difficult to conduct, which makes them rare. Adoption records are confidential and difficult to access. This makes finding the biological parents of an adopted child, which is necessary to conduct these studies, hard. The number of adoptions in the United States has decreased in the last few decades as well. Instead, more adopted children are coming from overseas, which makes identifying biological parents for research purposes even more difficult.

Twin studies: Twin research is an informative approach for understanding the genetic and environmental influences affecting behavioral, physical, and medical trait. Twin studies also help solve a scientific mystery. Twin studies compare behaviors of identical twins to one another. Think back to the study Jane was trying to conduct. A subject may develop an addiction to alcohol from their biological parents, regardless of their adoptive parents discouraging alcohol use. This may lead Jane to conclude that the subject's alcoholism developed as a result of genetic factors. However, this points out a flaw in adoptive studies. There are other environments that influence behavior besides how they are raised. Many children spend almost as much time at school with their teachers and friends as they do with their parents. This subject could have easily fallen in with a peer group that abused alcohol, putting them in an environment that would make it more likely for them to develop alcoholism.

Twin Studies

Definition: Twin studies compare similarities between monozygotic (identical) twins, who share 100% of their genes, and dizygotic (fraternal) twins, who share on average 50% of their segregating genes. This comparison allows researchers to estimate the heritability of traits—how much of the variation in a trait within a population is due to genetic factors.

Methodology:

Monozygotic Twins: These twins share nearly identical genetic makeup because they originate from a single fertilized egg that splits into two embryos. Any differences observed between them can often be attributed to environmental factors.

Dizygotic Twins: These twins develop from two separate eggs fertilized by two different sperm cells, similar to any other siblings. Comparing them with monozygotic twins helps researchers estimate genetic influences on traits.

Applications:

Heritability Estimates: Twin studies help quantify the extent to which genetic factors contribute to the variation observed in traits such as intelligence, personality traits, mental disorders, physical health, and more.

Gene-Environment Interactions: They also explore how genetic predispositions interact with environmental factors to influence traits and behaviors.

Adoption Studies

Definition: Adoption studies involve comparing adopted children to their adoptive parents (who provide the environmental influence) and their biological parents (who provide genetic influence but not shared environment). This comparison allows researchers to disentangle genetic and environmental influences on various traits.

Methodology:

Adopted Children: These children share genes with their biological parents but are raised in environments different from their genetic relatives, providing insight into genetic influences.

Adoptive Parents: These parents provide the environmental influence on the adopted child but do not share genes with them, allowing researchers to isolate environmental effects.

Applications:

Nature vs. Nurture: Adoption studies help researchers understand the relative contributions of genetics and environment to traits such as intelligence, personality, mental health outcomes, educational attainment, and more.

Gene-Environment Correlations: They examine how genetic predispositions might lead individuals to seek or create environments that reinforce their genetic tendencies.

Strengths and Limitations:

Strengths: Both twin and adoption studies provide opportunities to study the interplay between genes and environment, offering insights that can inform theories about human development and behavior.

Limitations: They can be complex to design and execute, especially concerning recruitment and data collection. Additionally, assumptions such as equal environments for twins or representativeness of adoptive families can impact results.

In summary, twin studies and adoption studies are instrumental in advancing our understanding of genetic and environmental influences on human traits and behaviors, providing complementary insights into the complex interactions that shape individual differences.

Advantages of Quasi-Experimental Design

Quasi-experimental designs have several advantages over other research designs, such as:

- **Greater external validity:** Quasi-experimental designs are more likely to have greater external validity than laboratory experiments because they are conducted in naturalistic settings. This means that the results are more likely to generalize to real-world situations.
- **Ethical considerations:** Quasi-experimental designs often involve naturally occurring events, such as natural disasters or policy changes. This means that researchers do not need to manipulate variables, which can raise ethical concerns.
- **More practical:** Quasi-experimental designs are often more practical than experimental designs because they are less expensive and easier to conduct. They can also be used to evaluate programs or policies that have already been implemented, which can save time and resources.
- **No random assignment:** Quasi-experimental designs do not require random assignment, which can be difficult or impossible in some cases, such as when studying the effects of a natural disaster. This means that researchers can still make causal inferences, although they must use statistical techniques to control for potential confounding variables.
- **Greater generalizability:** Quasi-experimental designs are often more generalizable than experimental designs because they include a wider range of participants and conditions. This can make the results more applicable to different populations and settings.

Limitations of Quasi-Experimental Design

There are several limitations associated with quasi-experimental designs, which include:

- **Lack of Randomization:** Quasi-experimental designs do not involve randomization of participants into groups, which means that the groups being studied may differ in important ways that could affect the outcome of the study. This can lead to problems with internal validity and limit the ability to make causal inferences.
- **Selection Bias:** Quasi-experimental designs may suffer from selection bias because participants are not randomly assigned to groups. Participants may self-select into groups or be assigned based on pre-existing characteristics, which may introduce bias into the study.
- **History and Maturation:** Quasi-experimental designs are susceptible to history and maturation effects, where the passage of time or other events may influence the outcome of the study.
- **Lack of Control:** Quasi-experimental designs may lack control over extraneous variables that could influence the outcome of the study. This can limit the ability to draw causal inferences from the study.
- **Limited Generalizability:** Quasi-experimental designs may have limited generalizability because the results may only apply to the specific population and context being studied.

FACTORIAL DESIGN

The independent variable (also known as "**factor**"), or cause, is what the researcher controls. It doesn't depend on any other variable in the study. The dependent variable, or effect, is the variable that changes in response to the independent variable and is what the researcher measures. A factorial research design is an experiment that has a minimum of two factors. These factors, also known as independent variables, are what the researcher controls. Factorial designs in which two or more independent variables are studied simultaneously in one experiment. factorial designs can also be called complex designs. It involves factorial combination of independent variables. Factorial combination involves pairing each level of one independent variable with each level of a second independent variable. This makes it possible to determine the effect of each independent variable alone (main effect) and the effect of the independent variables in combination (interaction effect).

Factorial Research Design: Main Effect

In a **factorial research design**, the main effect is an important feature to consider. The main effect refers to the effect of a factor on a dependent variable, averaged across the levels (factor subdivisions) of the other factors. Thus, each factor has one main effect.

This concept can be further illustrated with an example:

Researchers want to determine how the amount of sleep a person gets the night before an exam impacts performance on a math test the next day. But the experimenters also know that many people like to have a cup of coffee (or two) in the morning to help them get going.

For example, imagine that researchers want to test the effects of a memory-enhancing drug. Participants are given one of three different drug doses, and then asked to either complete a simple or complex memory task.

The researchers note that the effects of the memory drug are more pronounced with the simple memory tasks, but not as apparent when it comes to the complex tasks. In this 3×2 factorial design, there is an interaction effect between the drug dosage and the complexity of the memory task. So, the researchers decided to look at how the amount of sleep and caffeine influence test performance.

The researchers then decided to look at three levels of sleep (4 hours, 6 hours, and 8 hours) and only two levels of caffeine consumption (2 cups versus no coffee). In this case, the study is a 3×2 factorial design.

Example #1

Suppose a study on the effects of different types of online learning environments and study strategies on academic performance in college students is carried out. The study could use a 2×2 factorial design, with two independent variables (learning environment and study strategy) and two levels of each independent variable.

The learning environments could be synchronous online learning (i.e., live classes with real-time interaction), asynchronous online learning (i.e., pre-recorded lessons with discussion boards), or a combination. The study strategies are self-regulated learning (i.e., self-paced and self-directed study), collaborative learning (i.e., group work and peer feedback), or a combination.

Participants would be assigned groups: synchronous learning with self-regulated study, asynchronous understanding with collaborative study, synchronous and asynchronous learning with self-regulated study, or synchronous and asynchronous learning with collaborative research. The dependent variable would be the participants' academic performance, measured by their grades in a specific course or course.

By manipulating the levels of the learning environment and study strategy and measuring their combined and individual effects on academic performance, this study could provide valuable insights into the most effective approaches to online learning for college students

What is the purpose of factorial design?

A factorial design allows the researcher to examine the main effects of two or more independent variables simultaneously. It also allows the researcher to determine interactions among variables. This allows conclusions to be made and/or the testing of a hypothesis.

The simplest complex design is a 2 X2 design—two independent variables, each with two levels. The number of different conditions in a complex design can be determined by multiplying the number of levels for each independent variable (e.g., $2 \times 2 = 4$). More powerful and efficient complex designs can be created by including more levels of an independent variable or by including more independent variables in the design. In a 3 X 3 design there are two independent variables with three levels each, so there are nine conditions. In a 3X4X2 design there are three independent variables with three, four, and two levels, respectively, and a total of 24 conditions.

Types

It can be classified into several types. Based on the number of **independent variables** (factors) and levels used in the experiment. Some common types of it include:

1. **2×2 factorial design:** It involves two independent variables, each with two levels. It is popular in psychological research to investigate the effects of two factors on behavior or outcome.
2. **3×3 factorial design:** It involves three independent variables, each with three levels. It helps investigate the effects of multiple factors on behavior or outcome and can be particularly useful in medical research.
3. **Mixed factorial design:** This design involves at least one independent variable manipulated within subjects (i.e., each participant experiences all levels of the variable). And at least one independent variable between subjects (i.e., each participant experiences only one level of the variable).
4. **Nested factorial design:** This design involves one independent variable that is in alignment with another independent variable. For instance, a study on different types of therapy for depression might have one independent variable that represents the type of therapy (cognitive-behavioral

therapy, psychoanalytic therapy, etc.) and another independent variable that represents the therapist administering the treatment.

5. **Fractional factorial design:** This design involves testing only a subset of possible combinations of levels of the independent variables. This can be useful when resources are limited or when trying all possible combinations would be impractical.

MAIN EFFECTS

In factorial designs, there are three kinds of results that are of interest: main effects, interaction effects, and simple effects.

A main effect is the effect of one independent variable on the dependent variable—averaging across the levels of the other independent variable. Thus, there is one main effect to consider for each independent variable in the study. For example, shows a clear main effect of psychotherapy length. The longer the psychotherapy, the better it worked.

INTERACTIONS

There is an interaction effect (or just “interaction”) when the effect of one independent variable depends on the level of another. Although this might seem complicated, you already have an intuitive understanding of interactions.

Some examples of interactions is that the effect of receiving psychotherapy is stronger among people who are highly motivated to change than among people who are not motivated to change. This is an interaction because the effect of one independent variable (whether or not one receives psychotherapy) depends on the level of another (motivation to change).

Advantages and Disadvantages

The advantages are as follows:

1. **Ability to investigate multiple factors:** These allow researchers to investigate the effects of various independent on a dependent variable in a single experiment, which can save time and resources.
2. **Identification of main effects and interactions:** These enable researchers to identify the main products of each independent variable and any interaction effects between them, providing a more nuanced understanding of the relationships between variables.
3. **Increased statistical power:** By manipulating multiple independent variables, factorial designs can increase the statistical power of a study and improve the likelihood of detecting meaningful effects.
4. **Flexibility:** These adapt to various research questions and uses in multiple fields, including psychology, education, medicine, and engineering.

The disadvantages are as follows:

1. **Increased complexity:** Using multiple independent variables can interpret results more complexly, mainly when interaction effects are present.
2. **Increased sample size requirements:** A larger [sample size](#) is preferable to a more straightforward design with fewer independent variables to power such a design adequately.
3. **Potential for confounding:** The presence of interaction effects can make it challenging to determine which independent variable is responsible for observed effects, potentially confounding the results.
4. **Limited generalizability:** The specific conditions of it may not be generalizable to other contexts.

Small N design

Small-N Designs. One or only a few subjects tested ("single-subject designs"). Subjects not put into groups, but run as individuals. Expts are long, allowing performance to stabilize over time. Data from Ss usually not combined but considered separately. Data analyzed visually with minimal use of inferential stats.

Researchers manipulate an independent variable in single-subject experiments; therefore, these designs allow more rigorous control than case studies. • In single-subject experiments, baseline observations are first recorded to describe what an individual's behavior is like (and predicted to be like in the future) without treatment. • Baseline behavior and behavior following the intervention (treatment) are compared using visual inspection of recorded observations.

The first stage of a single-subject experiment is usually an observation stage, or baseline stage. During this stage researchers record the subject's behavior prior to any treatment. Clinical researchers typically measure the frequency of the target behavior within a unit of time, such as a day or an hour. For example, a researcher might record the number of times during a 10-minute interview that an excessively shy child makes eye contact, the number of headaches reported each week by a migraine sufferer, or the number of verbal pauses per minute made by a chronic stutterer.

An example of a small n design is the ABA approach. In this type of intervention, a baseline measurement is taken before treatment is offered, and the treatment is eventually withdrawn. Types of small n design:

The A-B-A design, which is also called the **ABA design**, is a method of behavioral intervention often used in [Applied Behavior Analysis](#). This design utilizes a three-stage process that includes [baseline measurement](#) (A), treatment (B), and withdrawal (A). The first stage of baseline measurement entails measuring a certain behavior or trait. Researchers accomplish this before giving any treatment to group members. Next, they offer the treatment to participants. The duration of treatment can vary greatly, and researchers can take measurements during the treatment. Lastly, the treatment is withdrawn, and they take the final measurement to see if the treatment worked and if the target behavior is changed.

An example cited in one article is that of children asked to read a paragraph that included text only. The children were tested on their understanding of the information. Then, another paragraph including an illustration was given to the children to read. Again, they were tested to see if their level of understanding increased. Finally, they were given another paragraph that

contained only text and retested to see if their grasp of the information returned to the initial test results. Using the ABA design, the therapist can evaluate the effects of treatment related to baseline responding.

Issues with Baseline (ABA) Designs:

- Need to establish a stable baseline; often based on an a priori "stabilization criterion".
- Design requires the use of a reversible treatment. Without reversal (in 2nd A phase) data could reflect carry-over effects (raising internal validity issues).
- Data based on one subject
- For some (e.g., therapeutic) studies, it may be inappropriate or unethical to withdraw the treatment (e.g., phobias, addiction, autism, etc.). In those cases, a simple AB design may be used (although the internal validity of an AB is quite low).
- For external validity replication with additional subjects

Extensions of Baseline Designs ABAB Design: Often preferred to ABA because (a) Two reversals occur (increasing interval validity) and (b) treatment is left in place. Data from Hall et al. (1968) examining study behavior in a disruptive 3rd grader as a function of "special attention". Note double reversal (increased internal validity); but also, failure to recover original baseline (carry-over?) □ ABA designs can also be extended to study multiple treatments in one person: e.g., 3 treatments could be studied in an ABACADA Design. Treatments can be combined (A-B-BC-A-BC; see text). Placebo effects can also be examined (A-A 1-B-A 1-B). Etc.

Multiple baseline design: Sometimes the ABA design is inappropriate for studies; in these cases, the **multiple baseline design** may be more suitable for these studies or interventions. One example of this is with measuring multiple behaviors. Although researchers use the multiple baseline design to measure multiple behaviors, they only focus on the change in the target behavior. Since the goal is to change one specific behavior, measuring a multitude of behaviors allows researchers to see if only the targeted behavior is the one that changed after a treatment is applied.

Multiple Baseline Designs □ Multiple Baseline Designs examine IV (treatment) effects across multiple Ss, behaviors, or situations by triggering the treatment at different points in time. Figure shows multiple Ss; can you extend to multiple behaviors or contexts? □ Separate baselines are established for different Ss, behaviors, or contexts. □ Treatments are then triggered at different times. □ If treatment is withdrawn, reversals can suggest high interval & external validity. □ Very useful in therapeutic & behavioral mod situations.

The multiple-baseline design also makes use of baseline and treatment stages, but not by withdrawing a treatment as in the ABAB design. As the name suggests, researchers establish several baselines when using a multiple-baseline design. The multiple-baseline design demonstrates the effect of a treatment by showing that behaviors in more than one baseline change following the introduction of a treatment. One example of

Across different situations: the multiple-baseline design is to treat one person's behavior in different situations. In this case, the first step in the multiple-baseline design is to record behavior

(such as the aggressiveness of a child) as it normally occurs in several situations (such as at home, in the classroom, and at an after-school daycare facility). The researcher establishes the baseline frequency of the behavior in each situation (i.e., multiple baselines). Next the treatment is introduced in one of the situations (e.g., at home), but not in the other situations. The researcher continues to monitor behavior in all of the situations. A critical feature of the multiple-baseline design is that treatment is applied to only one baseline at a time. The behavior in the treated situation should improve; the behavior in the baseline situations should not improve. The next step is to apply the treatment in a second situation (treatment may continue in the first situation as well) but leave the third situation as a continuing baseline. Behavior should change only in the treated situation, not in the baseline situation. The final step is to administer the treatment in the third situation; again, the behavior should change when the treatment is administered in the third situation. The key evidence for the effectiveness of a treatment in the multiple-baseline design is the demonstration that behavior changes only when the treatment is introduced. There are several variations on the multiple-baseline design, depending on whether multiple baselines are established for different individuals, for different behaviors in the same individual, or for the same individual in different situations.

In the multiple-baseline design across individuals, baselines are first established for different individuals. When the behavior of each individual has stabilized, an intervention is introduced for one individual, then for another individual, later for another, and so on. As in all multiple-baseline designs, the treatment is introduced at a different time for each baseline (in this case, for each individual). If the treatment is effective, then a change in behavior will occur immediately following the application of the treatment in each individual. An example of the use of a multiple-baseline design across individuals comes from the field of sports psychology. Allison and Ayllon (1980) were interested in evaluating the effectiveness of a coaching method that involved several behavioral techniques on the acquisition of specific football, tennis, and gymnastic skills. Although they found that the method was effective for each sport, we will describe their test of the effectiveness of behavioral coaching for the acquisition of a football skill. The participants in this experiment were second-string members of a citywide football program chosen because they “completely lacked fundamental football skills”

Multiple baseline design across behaviors: A second type of multiple-baseline design involves establishing two or more baselines by observing different behaviors in the same individual, a multiple baseline design across behaviors. A treatment is directed first at one behavior, then at another, and so on. Evidence for a causal relationship between treatment and behavior is obtained if performance changes for each behavior immediately after the treatment is introduced. For example, Gena, Krantz, McClannahan, and Poulson (1996) attempted to teach several different socially appropriate affective behaviors to youths with autism. As the researchers noted, children with autism often show inappropriate affective behaviors, which limit their opportunities to communicate effectively with others and to develop interpersonal relationships. Treatment included verbal praise and tokens (exchangeable for rewards) that were delivered contingent on appropriate affective responses in three or four different behavior categories. Target behaviors were selected from among the following: showing appreciation, talking about favorite things, laughing about absurdities, showing sympathy, and indicating dislike. Visual inspection of the behavioral records showed evidence for the effectiveness of the treatment for

each individual. As required in the multiple-baseline design, the different affective behaviors changed immediately after introduction of the intervention for that behavior.

The third major variation on the multiple-baseline design involves establishing two or more baselines for an individual's behavior across different situations, a multiple-baseline design across situations. For example, as we described when we introduced the multiple-baseline design, a researcher might establish baselines showing the frequency of a child's aggressive behavior at home, in the classroom, and at an after-school daycare facility. As with other variations of this design, the treatment is applied at different times and the behavioral records are examined to determine whether behavior changes systematically with the introduction of treatment.

MULTIPLE PROBE DESIGN • Analyzes relation between independent variable and acquisition of skill sequences • Instead of simultaneous baselines, probes provide basis for determining if behavior change has occurred prior to intervention • Appropriate for analyzing a shaping program • Need to be careful conducting the analysis **Changing Criterion Designs:**

Definition: An experimental design where the initial baseline phases are followed by a series of treatment phases consisting of successive and gradual changing criteria for reinforcement or punishment.

Example in clinical context: A behavior analyst wants to assess how a client's behavior changes when they provide reinforcement for every five responses per minute, then ten responses per minute and so on. The criterion increases as the client demonstrates stable states of responding. A behavior analyst is consulting with a client who wants to decrease the number of cigarettes they smoke per day with the goal of quitting. The client currently smokes 16 cigarettes per day. The first criterion the behavior analyst sets before the client can earn reinforcement is 13 cigarettes per day, to 10, seven, five and one. The criteria decrease as the client demonstrates stable states of responding.

Multielement/Alternating Treatments Design

Definition: An experimental design where two or more conditions are presented in rapidly alternating succession independent of the level of responding and the effects on the target behavior.

EXAMPLE: A behavior analysts is comparing two treatments with a client on the response rate of their aggressive behavior. The behavior analyst conducts a multielement/alternating treatments design on Treatment A and Treatment B. Treatment A did not appear to have an effect on the aggressive behavior, but Treatment B showed a sharp decrease in aggressive behavior.

Advantages of small N design:

One of the major advantages of the small-N experimental design is that it allows researchers to observe the effects of an intervention in a more controlled and systematic way than other research designs. By limiting the number of participants, researchers are able to observe the effects of the intervention in a more controlled environment. Additionally, the small-N experimental design allows researchers to examine the effects of an intervention over a relatively short time period. This is particularly useful for interventions that are expected to have a rapid effect on the participants.

Limitations:

the small-N experimental design also has some limitations. One of the most significant is that the small sample size can limit the generalizability of the findings. Additionally, because the intervention is administered to a single group, it can be difficult to separate the effects of the intervention from other factors that may have an influence on the results. Finally, the small-N

experimental design can be difficult to replicate, as it requires the same intervention to be applied to the same group of individuals multiple times.

Classical,

After-only research design: The After-Only Research Design gets its name from the fact that the dependent variable is measured only once after the experimental treatment. In other words, the posttest is administered once to the experimental group and the control group (see Figure 4.5). It shows an experiment in which the researcher is attempting to show the effectiveness of the inductive method in improving the science problem skills of 17-year-old secondary school students. The sample was drawn from a population and randomly assigned to the experimental and control group. The experimental group were taught science using the inductive approach while students in the control group were not taught using the inductive approach. Instead students in this group were taught the same science content using the traditional didactic approach (Âchalk and talkÊ method)

Before-After : Also known as the pre-test/post-test design. The Before-After Research Design is perhaps the best example of a true research design that incorporates both an experimental and control group to which the subjects are randomly assigned (see Figure 4.12). This research design is a good experimental design because it does a good job of controlling for extraneous factors such as history, maturation, instrumentation, selection bias and regression to the mean. How is this done? Any history events (e.g. certain events subjects may have been exposed to) that may have produced a difference in the experimental group would also have produced a difference in the control group. Here it is assumed that subjects in both groups have experienced the same set attitude

It is one of the most commonly used designs in evaluation studies

Examples

Impact of administrative restructuring on the quality of services provided by an organisation

Effectiveness of a marriage counselling service.

The effect of an advertisement on the sale of a product

The impact of increased funding on the quality of teaching in universities.

The impact of maternal and child health services on the infant mortality rate

The impact of incentives on the productivity of employees in an organisation

Advantages: The main advantage is that it can measure change in a situation, phenomenon, issue, problem or attitude. Most appropriate design for measuring the impact or effectiveness of a program

Disadvantages

- More expensive and more difficult to implement.

It also requires a longer time to complete, particularly if you are using an experimental design.

- Time lapse between the two contacts may result in attrition in the study population

Main limitations of this design, in its simplest form, is that as it measures total change, you cannot ascertain whether independent or extraneous variables are responsible for producing change in the dependent variable. Sometimes the instrument itself educates the respondents. This is known as the reactive effect of the instrument

No control group: one-Shot Design For example, you want to determine whether praising primary school children makes them do better in arithmetic (see Figure 4.1). You measure arithmetic achievement with a test. To test this idea, you choose a class of year 4 pupils and increase praising of children and you find that their mathematics performance significantly improved. X (praise) O (scores on a mathematics test) Figure 4.1: One-shot design You conclude that praising children increases their mathematics score. This design is weak for the following reasons: 1. Selection Bias: It is possible that the pupils you selected as subjects were already good in mathematics. 2. History: The school had organised a motivation course on mathematics for year 4 pupils. So, it is possible it might influence their performance.

crossover, the *crossover design* compares two or more interventions by randomly assigning each participant to receive the interventions being tested in a different sequence. Once one intervention is completed, participants are switched to another intervention. For example, in a two-by-two crossover design, each participant randomly receives one drug for one period of time and then another drug for a second period of time, with the administration of each drug separated by a washout period (i.e., a period of time during which the first drug is cleared from the body before the second drug is administered). With this type of study, each participant serves as his or her own control.

Crossover Trial

In a crossover trial, patients start off on one arm (i.e., receive one intervention) during one period and then switch to another arm (i.e., receive another intervention) for the next period. Every time a patient switches arms, the patient “crosses over” to another arm. Depending on the study, a patient may crossover multiple times. In the racing analogy, a crossover is like switching shoes. The crossover trial is the most common within-patient design.

The simplest crossover design is the two-period crossover: during the first period, patients receive one of two treatments and during the second period patients receive the other treatment (e.g., give half the patients an AB sequence and the other half a BA sequence). Figure 6.3 shows a schematic of a two-period crossover study.

Crossover Study Analysis

In a crossover study, the first step in analysis is an interaction analysis to determine whether there is a difference between the two periods. If there is no difference, then the periods can be combined. Otherwise, only the first period can be used.

For example, let's consider a study where the first period group A receives placebo and group B receives active drug, and then in the second period group A receives active drug and group B receives placebo. The primary endpoint is reduction in the blood pressure. The first step is to determine whether the periods show an interaction. If there is not, then the group A period one data can be combined with group B period two and group A period two can be combined with group B period one.

Advantages: There are several advantages to this trial design, including a reduction in the number of participants required to achieve a statistically significant result and the ability to control for patient specific effects. This design can also be useful for studying a patient's response to short periods of therapy, particularly for chronic conditions in which the initial evaluation of treatment efficacy is concerned with the measurement of short-term relief of symptoms (Pocock, 1984).

Limitations: A criticism of this design is that the effects of one intervention may carry over into the period when the next intervention is given. Crossover studies cannot be done if the effects of the interventions are irreversible (e.g., gene therapy or surgery) or the disease progression is not stable over time (e.g., advanced cancer). Additional problems with crossover studies occur if participants withdraw from the study before they receive both interventions or the outcomes are affected by the order in which the interventions are administered (Senn, 1993).

Crossover designs are occasionally used in psychological studies because of the opportunity to use each patient at least twice and because of the probability that the component of the variance within individual patients is smaller than between patients (Matthews, 1995).

Solomon Four-group: The **Solomon four-group design** is a research method developed by Richard Solomon in 1949. It is sometimes used in social science, psychology and medicine. It can be used if there are concerns that the treatment might be sensitized by the pre-test. It include 4 groups 2 experiment group and 2 control groups. Random assignment to 4 groups. Only experiment 1 and control 1 receive pre-test. Post-test to all 4 groups. Treatment is applied to both experimental group.it minimize the threat to internal and external validity. Also minimize reactive effect of pre-test.

The four groups in the Solomon design

Group Pre-intervention Intervention Post-intervention

1	Test	Treatment	Test
2	Test	Control	Test
3	No test	Treatment	Test
4	No test	Control	Test

The first two groups receive the evaluation test before and after the study, as in a normal twogroup trial. The second groups receive the evaluation only after the study. The effectiveness of the treatment can be evaluated by comparisons between groups 1 and 3 and between groups 2 and 4. In addition, the effect of the pre-treatment evaluation can be calculated by comparing the control group who received the pre-treatment evaluation with those who did not (groups 2 and 4).

Various statistical treatments for the Solomon four-group design have been put forward, including Stouffer's Z and Monte Carlo

Example				
The Solomon Four- Group Design				
100 randomly selected teachers	R 25 teachers (Exp-1)	O Pretest: Faculty Self-Awareness Questionnaire	X Dance for Peace Training- Workshop	O Posttest: Faculty Self-Awareness Questionnaire
	R 25 teachers (Con-1)	O Pretest: Faculty Self-Awareness Questionnaire	C No training	O Posttest: Faculty Self-Awareness Questionnaire
	R 25 teachers (Exp-2)		X Dance for Peace Training- Workshop	O Posttest: Faculty Self-Awareness Questionnaire
	R 25 teachers		C No training	O Posttest: Faculty Self-Awareness

Limitations of the Solomon four-group design

The Solomon four-group design is not used a lot in practice because of its high cost, since compared to a standard design with 2 groups it requires twice the sample size, time, materials, resources, and personnel/work.

